

Auteur: Victoria Mazur
Studierichting: Informatica
Oefeningenreeks 7A nr. 10.16

$$f(u, v) = (2u^2 + 3v^2) \cdot e^{-u^2-v^2} = 2u^2 \cdot e^{-u^2-v^2} + 3v^2 \cdot e^{-u^2-v^2}$$

$$\begin{aligned}\frac{\partial f}{\partial u}(u, v) &= 4u \cdot e^{-u^2-v^2} + 2u^2 \cdot (-2u) \cdot e^{-u^2-v^2} + 3v^2 \cdot (-2u) \cdot e^{-u^2-v^2} \\ &= (4u - 4u^3 - 6uv^2) \cdot e^{-u^2-v^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial f}{\partial v}(u, v) &= 2u^2 \cdot (-2v) \cdot e^{-u^2-v^2} + 6v \cdot e^{-u^2-v^2} + 3v^2 \cdot (-2v) \cdot e^{-u^2-v^2} \\ &= (-4u^2v + 6v - 6v^3) \cdot e^{-u^2-v^2}\end{aligned}$$

$$\Rightarrow Df(u, v) = \begin{pmatrix} (4u - 4u^3 - 6uv^2) \cdot e^{-u^2-v^2} & (-4u^2v + 6v - 6v^3) \cdot e^{-u^2-v^2} \end{pmatrix}$$

References